Newton's second law, for air.

Every tiny parcel of air obeys $F=ma$. Written for a fluid, it becomes the Navier–Stokes equation: the one the four screens are solving.

m mass $\qquad$ a acceleration

$$\rho\left(\frac{\partial v}{\partial t}+(v\cdot\nabla)v\right)$$

Density
Mass of one cubic metre of air: 1.2 kg

Speeding up in time
Zero once the flow has settled, as in our runs

Speeding up in space
The air changes speed as it moves to a new place around the shape

F forces

$$=-\nabla p+\mu\nabla^2 v$$

Pressure push
Air is shoved from high pressure towards low pressure

Friction
Each layer of air drags on its neighbours, and on the surface

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$\nabla\cdot v=0$ **And no air goes missing.** As much flows into any small box as flows out of it.